

ERZURUM BOLGE, Turkey

WMO: 170950

Lat: 39.9058N

Lon: 41.2544E

Elev: 6102

StdP: 11.73

Time Zone: 3.00 (E03)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.1	-7.7	-18.6	2.5	-12.0	-13.1	3.4	-7.5	30.6	26.6	24.4	23.0	0.4	60	0.568

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	28.5	87.7	57.2	85.5	57.0	82.9	57.3	62.9	79.8	61.3	78.1	59.6	76.4	6.7	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.1	84.2	71.8	54.4	79.1	69.9	52.9	74.9	68.3	31.7	79.6	30.4	78.4	29.1	75.9	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.7	19.9	15.9	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.5	16.6	20.7	30.4	43.0	51.4	59.6	68.0	69.1	59.1	47.8	32.2	23.2
	DBStd	19.48	10.90	10.34	10.49	6.84	5.09	5.51	4.50	4.67	6.23	5.85	9.16	11.04
	HDD50	4217	1036	821	608	222	45	1	0	0	8	112	533	831
	HDD65	8121	1501	1241	1072	661	422	178	21	19	193	534	983	1296
	CDD50	1859	0	0	0	11	88	288	559	591	280	42	0	0
	CDD65	292	0	0	0	0	0	15	115	146	16	0	0	0
	CDH74	4217	0	0	0	0	14	282	1602	1915	403	1	0	0
	CDH80	1271	0	0	0	0	0	35	504	664	68	0	0	0
Wind	WSAvg	4.2	3.3	3.0	4.3	5.2	5.2	4.7	5.2	4.7	4.0	4.2	3.0	3.0
Precipitation	PrecAvg	18.4	1.1	1.5	2.0	2.8	2.6	1.6	0.8	0.6	0.8	1.8	1.5	1.4
	PrecMax	21.9	3.3	4.4	3.7	6.5	4.7	3.1	1.9	2.0	2.6	3.7	3.0	3.0
	PrecMin	12.9	0.2	0.7	0.3	1.0	1.1	0.3	0.1	0.0	0.0	0.2	0.0	0.2
	PrecStd	2.6	0.6	0.7	0.8	1.1	1.0	0.8	0.4	0.4	0.6	1.0	0.8	0.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.3	43.5	56.3	67.7	75.7	83.6	91.0	90.6	84.4	72.2	55.8	43.6
		MCWB	32.4	36.8	42.1	52.5	53.5	58.5	58.5	55.8	52.8	49.9	40.7	36.3
	2%	DB	35.7	39.7	50.6	63.0	71.0	80.3	87.9	88.2	82.2	68.2	52.6	39.7
		MCWB	31.0	34.2	39.6	47.7	52.6	57.9	58.4	56.2	53.6	48.7	40.2	35.0
	5%	DB	33.4	37.3	47.3	59.7	67.3	77.1	85.3	86.2	78.9	64.1	48.7	37.7
		MCWB	29.7	32.9	38.1	45.9	51.2	56.2	58.5	56.3	53.4	48.3	38.5	34.1
	10%	DB	31.6	34.5	43.6	56.1	64.2	73.7	82.5	84.1	75.3	59.9	45.0	35.8
		MCWB	28.2	31.3	36.7	44.5	50.2	55.0	59.0	56.9	53.0	46.7	37.1	32.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.3	37.4	44.2	53.8	57.7	62.8	66.7	62.9	59.6	53.5	44.1	37.9
		MCDB	36.5	42.8	51.2	64.0	71.2	78.2	84.1	81.2	72.5	67.0	50.0	41.4
	2%	WB	31.7	34.9	41.0	50.3	54.4	60.0	64.1	61.2	56.6	51.0	41.9	36.0
		MCDB	35.1	38.5	50.0	59.8	67.3	75.7	81.1	79.5	75.1	63.9	49.7	38.9
	5%	WB	30.1	33.2	38.7	47.8	52.6	58.3	62.4	59.8	54.9	49.2	39.8	34.5
		MCDB	33.4	36.4	45.5	56.7	64.4	73.5	79.3	78.6	74.1	60.9	47.3	37.1
	10%	WB	28.1	31.5	36.8	45.4	50.8	56.7	60.8	58.7	53.6	47.7	37.7	32.8
		MCDB	31.2	34.5	43.3	54.3	61.6	71.4	77.1	78.1	72.1	57.8	43.9	35.3
Mean Daily Temperature Range	5% DB	MDBR	17.1	18.2	17.9	20.9	21.8	25.0	28.2	28.5	28.9	21.6	18.3	14.7
		MCDBR	13.3	15.6	20.3	26.8	28.4	29.4	31.3	30.9	33.4	28.8	23.1	12.0
		MCWBR	9.7	11.2	12.2	15.6	14.8	12.6	12.3	10.5	13.6	14.4	13.2	8.3
	5% WB	MCDBR	12.3	14.8	18.5	25.3	26.1	28.2	29.7	29.1	32.6	25.3	20.0	10.2
MCWBR		9.3	11.2	11.9	18.2	15.3	14.7	14.4	12.1	16.0	14.0	11.9	7.5	
Clear-Sky Solar Irradiance	taub	0.269	0.306	0.359	0.416	0.385	0.349	0.363	0.367	0.317	0.344	0.292	0.265	
	taud	2.242	2.116	2.063	2.031	2.221	2.369	2.350	2.318	2.442	2.303	2.405	2.338	
	Ebn at Noon	284	286	281	272	283	293	288	284	294	270	275	278	
	Edh at Noon	30	40	47	53	44	38	39	39	32	33	26	25	
All-Sky Solar Radiation	RadAvg	793	1157	1472	1552	1811	2255	2303	2060	1693	1101	751	626	
	RadStd	83	95	145	160	136	128	120	87	100	88	103	100	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	+1.15	-0.81	N/A	N/A	N/A	+160	+67

Nomenclature: See separate page